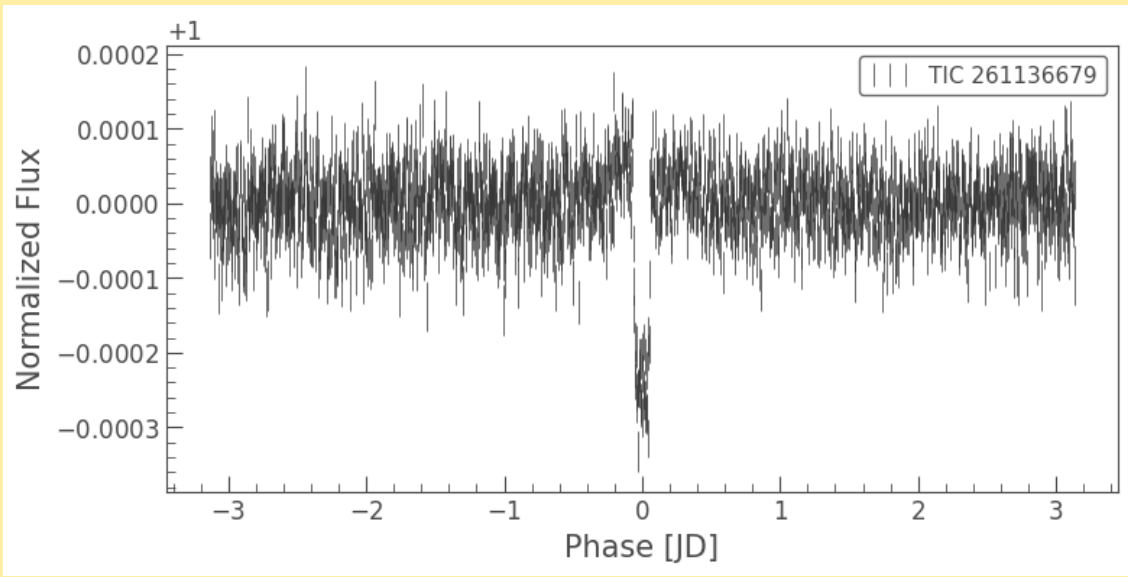


Exoplanet Detection via Kepler Data: A Comparative Study of Quantum and Classical Machine Learning Models

INTRODUCTION

What?

Exploring exoplanets has become a cornerstone of modern astrophysics. My research compares the design and implementation of classical and quantum machine-learning techniques to detect exoplanets within the Kepler Objects of Interest (KOI) Cumulative Dataset from the NASA Exoplanet Archive. Kepler primarily uses the transit method, where exoplanets reveal themselves by dimming a star's light as they pass in front of it. If this dimming occurs at regular intervals and lasts a fixed, repeated time, it likely indicates an orbiting object. Machine learning models can detect these subtle, repeating patterns even in weak or noisy signals. Plotting stellar brightness over time reveals a light curve with periodic dips, each indicating a planetary transit. I used the Lightcurve Python module to analyze light curves from satellites and, after extensive data manipulation, obtained the transit light curve for a confirmed exoplanet named Pi Mensae.



Why?

Missions like Kepler and TESS observe hundreds of thousands of stars, generating terabytes of data which is quite noisy, with irregularities caused by instruments, starspots, or cosmic rays. Machine learning handles this scale far more efficiently than manual analysis or simple rule-based filters. Also, ML models don't need explicit equations to detect transit shapes—they learn from examples. Machine learning classifiers (like the ones used in my research) can learn to distinguish between real planets and imposters using labeled training data. Classical machine learning still faces limitations when it comes to processing extremely high-dimensional, noisy, and entangled information. This is where Quantum Machine Learning becomes an exciting and valid frontier to explore. It introduces a fundamentally new approach and leverages the unique capabilities of quantum computing, which may process high-dimensional data more efficiently than classical methods. One of the primary objectives of my approach is to develop quantum algorithms for machine learning, encompassing both the adaptation of classical machine learning models and the creation of new models inspired by quantum computing principles.

CLASSICAL MACHINE LEARNING MODEL

Data Preprocessing:

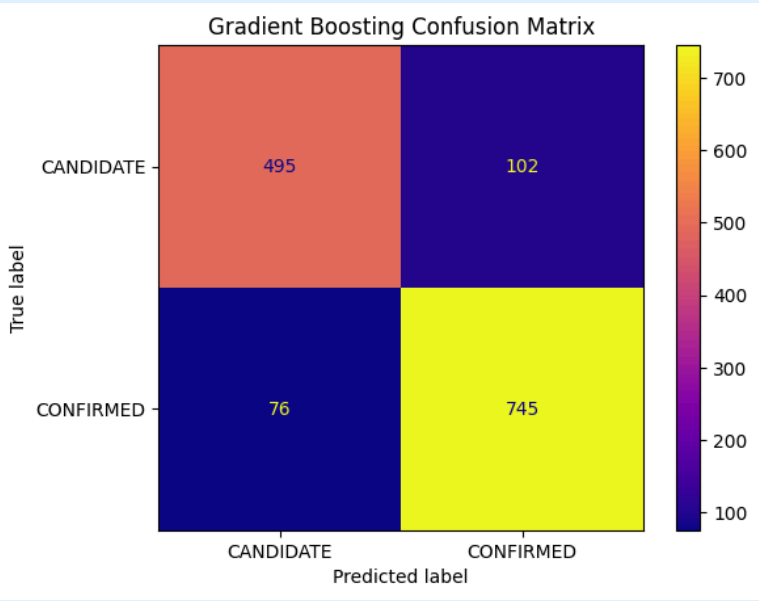
I started off by data preprocessing, which involved dropping rows/columns with missing values, one-hot encoding categorical features, splitting data into train and test sets, and scaling data using StandardScaler(). I turned it into a binary classification problem and split the data into 70% training and 30% testing.

Training the model:

Now on the preprocessed data, I trained four ML algorithms, which include: Logistic Regression, Neural Network — MLP Classifier, Random Forest Classifier and Gradient Boosting Classifier.

Analysis:

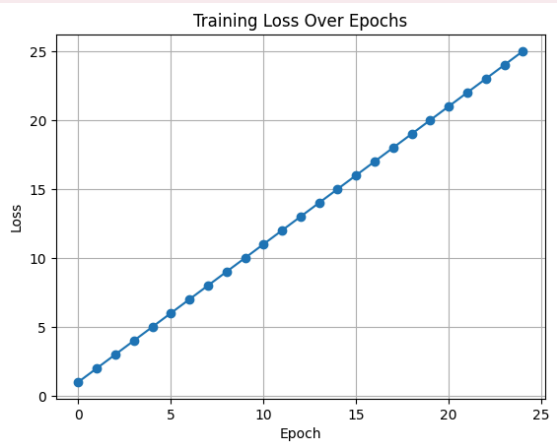
I then evaluated these models based on accuracy, precision, F1-score, and recall metrics. After analyzing the performance metrics, it was concluded that the Gradient Boosting Classifier was the best choice. To evaluate my model further, I visualized a confusion matrix of the Gradient Boosting Classifier as a heatmap, to show how many samples were correctly vs. incorrectly classified. The confusion matrix breaks down the predictions into correct and incorrect categories, helping one go beyond accuracy to evaluate how the model behaves per class.



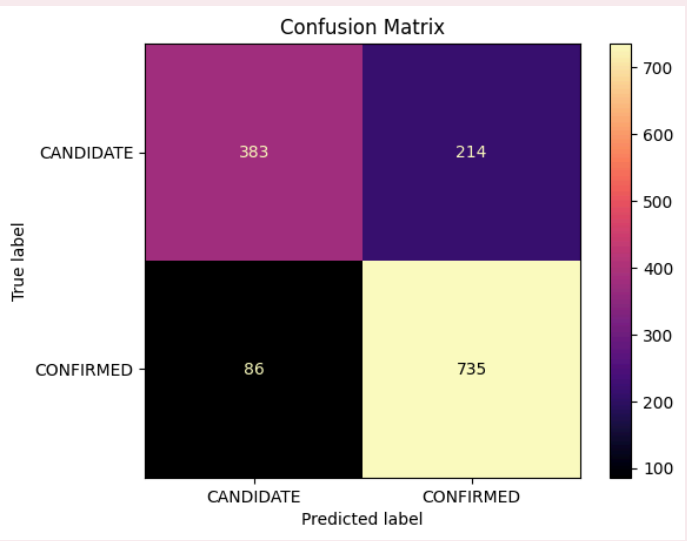
QUANTUM MACHINE LEARNING MODEL

I built a hybrid quantum-classical neural network using PennyLane and PyTorch, structured for binary classification.

I started the preprocessing by dropping irrelevant columns, filtering the dataset to include only two target classes and encoded them to: 'CANDIDATE' → 0 and 'CONFIRMED' → 1, filling missing values using the mean for numerical columns and mode for categorical ones, one-hot encoding categorical features, standardizing all feature values to have zero mean and unit variance, padding or truncating the resulting feature vectors to length 64, which corresponds to 2^6 — the required size for amplitude embedding on 6 qubits and then normalizing each 64-D vector to unit length to satisfy quantum state requirements. After this, I encoded classical data into a quantum state, allowing it to be trained with classical optimizers. My model consists of a quantum layer (the QNode wrapped by TorchLayer) which outputs 6 real values, a fully connected linear layer (nn.Linear) that maps these 6 values to a single logit. And a sigmoid activation function to output a probability score between 0 and 1, suitable for binary classification. To avoid a shape mismatch error during training, I converted the NumPy arrays to PyTorch tensors and wrapped them into a TensorDataset for use with DataLoader. I then trained for 25 epochs using mini-batch gradient descent (batch size = 8). And for each batch, I first cleared previous gradients, then passed inputs through the hybrid model, computed binary cross-entropy loss between predictions and labels, performed backpropagation to compute gradients across both classical and quantum parameters and updated the model using the Adam optimizer with a learning rate of 0.01. After this, I logged the epoch-wise total loss, observing its decrease over time to monitor convergence.



Lastly, I evaluated my model based on accuracy, precision, F1-score, and recall metrics. And furthermore, visualized a confusion matrix as a heatmap to show how many confirmed exoplanets and candidates were correctly identified by the model and how many were not.



COMPARISON: Performance Metrics

	Accurac y	Precision	F1 Score	Recall
Classical ML Model	87.87%	88.22%	89.70%	91.23%
Quantum ML Model	78.00%	76.65%	82.43%	89.16%

CONCLUSION

After a thorough analysis of both classical and hybrid quantum-classical machine learning models, I conclude that, based on the methods and implementation used in my project, the quantum approach **does not** demonstrate a clear performance advantage. While the quantum model successfully integrated amplitude embedding and variational circuits, its classification accuracy and learning dynamics did not significantly outperform the classical baseline. Further exploration with alternative quantum architectures, larger datasets, or more advanced encoding techniques may be necessary to fully assess the potential benefits of quantum machine learning in detecting exoplanets using Kepler's Data.



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